

Discovery data (GEO)

GSE130437 MCF7 palbo-R
GSE222367 MCF7 / T47D palbo-R
GSE229235 ER+ PDX palbo-R
49 resistant vs 25 sensitive

Differential expression

Moderated t per dataset
Robust rank aggregation
378 robust genes
(264 up, 114 down)

Function and network

GO / KEGG / Hallmark
Preranked GSEA
STRING v12 network
14 interferon hub genes

Machine learning

LASSO · SVM-RFE · Boruta
Gradient boosting
17-gene core signature
Nested leave-one-dataset-out

Clinical validation

TCGA-BRCA ER+/HER2- (n = 643)
METABRIC ER+/HER2- (n = 1396)
LASSO-Cox risk score
Nomogram · calibration · DCA

On-treatment biopsies

NeoPalAna trial (GSE93204)
C1D1 vs C1D15, paired
E2F, interferon and
resistance scores

Structure-based drug repurposing against STAT1

STAT1 SH2 phosphotyrosine subsite (PDB 1BF5, biological dimer) · redocking validation
1,176 approved drugs (ChEMBL) · AutoDock Vina two-stage screen · triplicate re-docking
Drug-likeness and interaction analysis of the top-ranked compounds